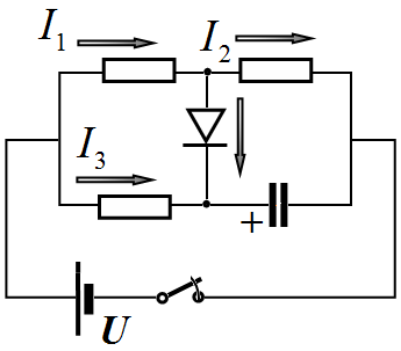
	(15) $T = 2\pi\sqrt{\frac{h_0}{g}}$	0,2	
합계			**3,0**

문제 B (4.0점)



I_k k (
), q_k
, q
, Q .

,
.

$I_2 = 0$

$I_0 = I_1(0) = U/R = 1$ mA. (1)

,

: $U_1 = U_2 = U/2$. .
: ,
,
: $U_1 = U_2 = U/2$. ,
:

$Q = CU/2$. (2)

:

$q_1 = q + q_2$, (3)
$q_3 + q = Q$. (4)
$I_1R = I_3R$,
$q_1 = q_3$, (5)
$U = I_1R + I_2R$. (6)

0 τ :

$U\tau = q_1R + q_2R$. (7)

$$q = \frac{1}{3}CU\left(1 - \frac{\tau}{RC}\right) = 179\ \mu\text{Cl}. \quad (8)$$

평가 항목	배점
(1) $I_0 = I_1(0) = U/R = 1$ mA	0.5
$I_0 = 1$ mA	0.1
(2) $Q = CU/2$	0.5
(3) $q_1 = q + q_2$	0.5
(4) $q_3 + q = Q$	0.5
(5) $q_1 = q_3$	0.5
(6) $U = I_1R + I_2R$	0.2
(7) $U\tau = q_1R + q_2R$	0.5
(8) $q = \frac{1}{3}CU\left(1 - \frac{\tau}{RC}\right)$	0.5
$q = 179\ \mu$Cl	0.2
합계	**4.0**